

Predictive Analysis on Airline Customer Satisfaction Using Supervised Learning

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Executive Summary

This report presents a predictive analytics study on airline customer satisfaction aimed at helping operators identify key operational drivers, optimize service delivery, and mitigate passenger churn in a highly competitive industry. Using a publicly available Kaggle dataset comprising 129,880 passenger records across 23 features, the data was thoroughly cleaned, preprocessed, and transformed—including power scaling for skewed continuous variables and the engineering of a novel binary predictor, "In Business Class". Three supervised classification algorithms were trained and evaluated: Logistic Regression as a baseline, alongside two ensemble approaches, Random Forest and LightGBM. Following hyperparameter tuning via randomized search, LightGBM was selected as the optimal operational model owing to its robust 5-fold cross-validation performance, superior computational efficiency, and strong discriminative ability. Evaluated on the test dataset, the ensemble models achieved exceptional classification accuracy exceeding 93.5%, with high precision and ROC-AUC scores near 0.985. Global SHAP interpretation revealed that inflight entertainment, seat comfort, ease of online booking, and passenger seating class are the primary determinants of customer satisfaction. Consequently, airlines should prioritize investments in cabin comfort upgrades and seamless digital touchpoints to maximize customer retention.

Keywords: Customer satisfaction, supervised learning, airline

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Introduction

Background

At the heart of any enterprise in the service industry is satisfying its customers. In the highly competitive airline industry, passenger satisfaction is a critical determinant of success, particularly in retaining passengers and fostering long-term brand loyalty (Eshagi et al., 2024). Understanding the distinct factors that drive customer experiences allows airlines to improve service quality and address specific consumer needs.

This problem is of significant importance because of the direct relationship between passenger satisfaction and business sustainability. Loyal customers represent the majority of passengers, and they generally exhibit higher rates of satisfaction compared to disloyal ones. By applying supervised learning techniques to model customer satisfaction, airline operators can predict passenger feedback and identify key drivers of positive experiences, such as seat comfort and inflight entertainment.

Problem Statement

Airlines face significant challenges in retaining passengers and fostering long-term brand loyalty due to intense industry competition. Understanding the factors driving customer satisfaction is essential for business sustainability, yet operators often lack the ability to accurately predict passenger feedback and identify the specific service attributes—such as seat comfort, inflight entertainment, and online services—that most influence satisfaction. This study addresses these challenges by developing and comparing supervised learning models, specifically Logistic Regression, Random Forest, and LightGBM, to predict customer satisfaction and identify the key service drivers that can be leveraged to improve service quality.

Objectives

General Objective

The primary objective of this study is to develop a robust predictive model that accurately classifies airline customer satisfaction and identifies the key service quality attributes that drive positive passenger experiences. By training and comparing supervised machine learning models on an airline customer satisfaction dataset, this research aims to provide airline operators with data-driven insights to optimize service delivery and enhance passenger loyalty.

Specific Objectives

- To clean and preprocess the Airlines Customer Satisfaction dataset, addressing missing values, duplicate entries, and categorical variable encoding.
- To perform feature scaling on skewed numerical variables using appropriate quantile and fractional power transformation techniques.
- To engineer a new binary feature, "In Business Class," to isolate and evaluate the impact of premium seating on overall customer satisfaction.
- To implement feature selection techniques, including Mutual Information and a Borda Count ranking system from tree-based model importances, to select the top predictive features.
- To train, hyperparameter tune, and evaluate three supervised learning classification models: Logistic Regression, Random Forest, and LightGBM.
- To compare the performance of the classification models using key evaluation metrics, such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- To perform model interpretation using SHAP analysis to determine global and local feature importance and provide business recommendations for airline service improvement.

Dataset Description

Dataset Source

The dataset used in this study is entitled Airlines Customer Satisfaction by Jana, S. (2020) obtained from the data science competition platform, Kaggle, on September 14, 2025. It has a total of 129,880 observations and 23 variables.

Variable Description

Of the 23 variables, there are 5 categorical variables, 4 numerical variables, and 14 ordinal/Likert scale variables (see Table 1). Our target variable is a binary categorical variable called *satisfaction* with values either *satisfied* or *dissatisfied*.

Column name	Values	Column name	Values
satisfaction	[satisfied, dissatisfied]	Inflight entertainment	Likert Scale (0-5)

Gender	[Male, Female]	Online support	Likert Scale (0-5)
Customer Type	[Loyal Customer, disloyal Customer]	Ease of Online booking	Likert Scale (0-5)
Age	Integer	On-board service	Likert Scale (0-5)
Type of Travel	[Personal Travel, Business travel]	Leg room service	Likert Scale (0-5)
Class	[Eco, Eco Plus, Business]	Baggage handling	Likert Scale (0-5)
Flight Distance	Integer	Checkin service	Likert Scale (0-5)
Seat comfort	Likert Scale (0-5)	Cleanliness	Likert Scale (0-5)
Departure/Arrival time convenient	Likert Scale (0-5)	Online boarding	Likert Scale (0-5)
Food and drink	Likert Scale (0-5)	Departure Delay in Minutes	Integer
Gate location	Likert Scale (0-5)	Arrival Delay in Minutes	Integer
Inflight wifi service	Likert Scale (0-5)		

Table 1. Dataset variable value information

Target Variable Analysis

Initial analysis on the target variable shows high class balance with 54.73% classified as *satisfied* while 45.27% are classified as *dissatisfied* (see Figure 1).

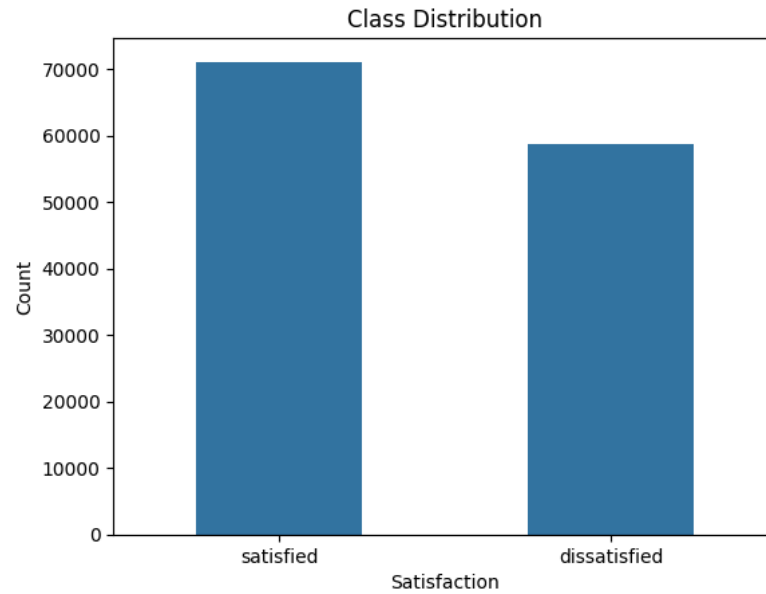


Figure 1. Target variable class distribution

Data Preprocessing

Missing Values and Duplicates

There are no duplicate values found in the dataset. However, upon further investigation, we discovered that only one feature called *Arrival Delay in Minutes* has null values which also happens to have a strong positive linear relationship (0.983) with *Departure Delay in Minutes* (see Figure 2). Given this observation, the former is excluded from the model.

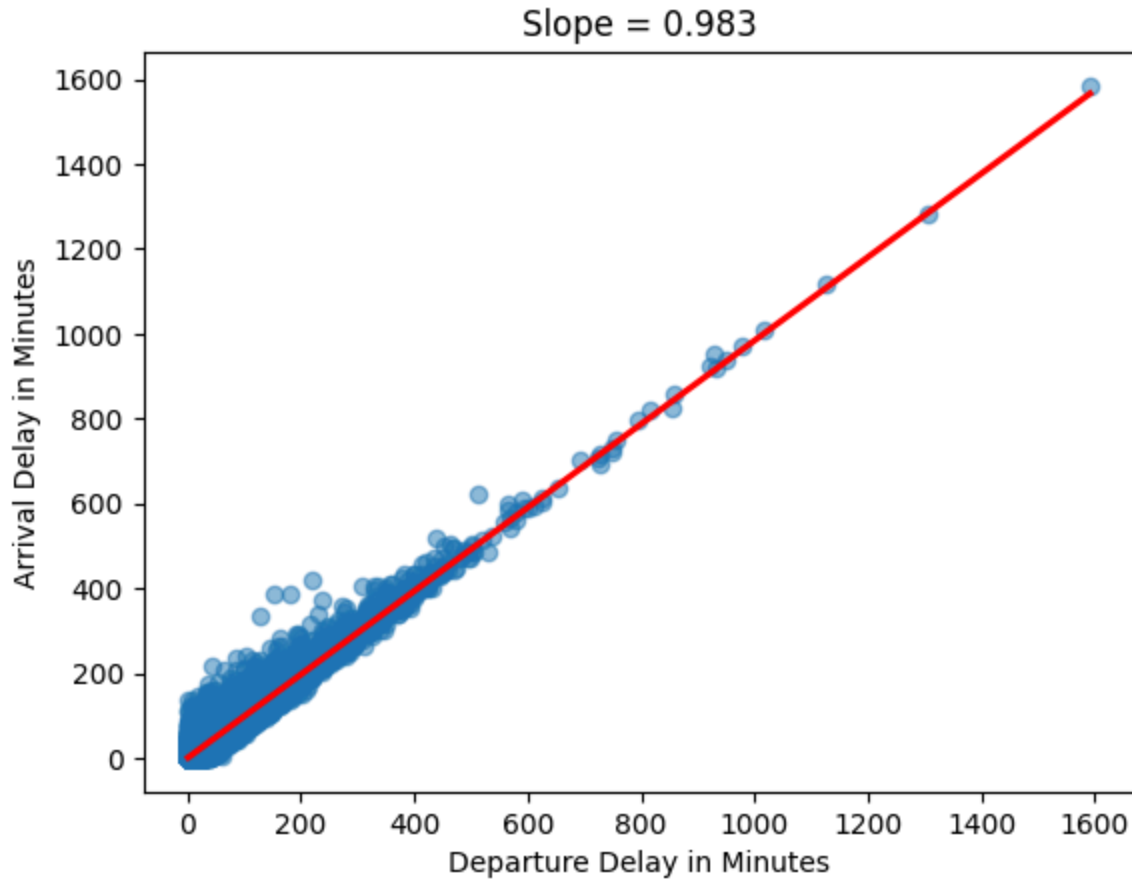


Figure 2. Scatter and regression plot of Departure Delay in Minutes vs Arrival Delay in Minutes

Feature Encoding

Categorical variables are encoded from object type to integers including the target variable. Only the *Class* feature is one-hot encoded since all other variables only have 2 unique values. Likert variables are already in their numerical form.

Feature Scaling

Scaling is necessary because we are using algorithms that perform significantly better when residuals and variables are normally and symmetrically distributed. Since numeric variables in the dataset have different levels of skewness (see Table 2), different techniques will be used in scaling them. For heavily skewed variables, the Quantile transformer is used while for those with relatively smaller skewness values, fractional power transformer is used.

Quantile transformer

This transformer maps numerical data to a uniform or normal distribution by transforming each feature independently to follow a specific quantile distribution. It is particularly effective for dealing with outliers and non-linear data distributions, making features more robust for machine learning models.

Fractional power transformer

The fractional power transformer works by applying a power function to each data point, which mathematically compresses the longer tail of a distribution to pull extreme values closer to the mean. This process stabilizes variance and makes the data more symmetric, making it particularly effective for slightly skewed distributions where the skewness is not severe enough to require more aggressive non-linear transformations.

Feature	Mean	Median	Standard Deviation	Skewness
Age	39.43	40	15.12	-0.003606
Flight Distance	1981.41	1925	1027.12	0.466748
Departure Delay in Minutes	14.71	0	38.07	6.821980
Arrival Delay in Minutes	15.16	0	38.60	6.641309

Table 2. Summary statistics of numerical variables

Exploratory Data Analysis

Numerical

According to Table 2, the numerical variables are right-skewed. Plots for *Departure Delay in Minutes* show large numbers of outliers and a long right tail. *Flight Distance* is also right-skewed although not as heavily as the previous two features. *Age* is the least skewed distribution and is very slightly left-skewed.

Course of Action:

- Since *Age* has a low absolute value of skewness, no transformation is needed.

- *Flight Distance* has a skewness score of 0.4667. Thus, we will use a fractional power transformation. We found 0.73 to be the optimal exponent which gives the transformed variable a skewness of 0.0009.
- We transform *Departure Delay in Minutes* using the Quantile transformer method due to its heavy skewness. After transformation, it now has a skewness value of 0.3117.
- After transforming all numerical values, the original features are dropped.

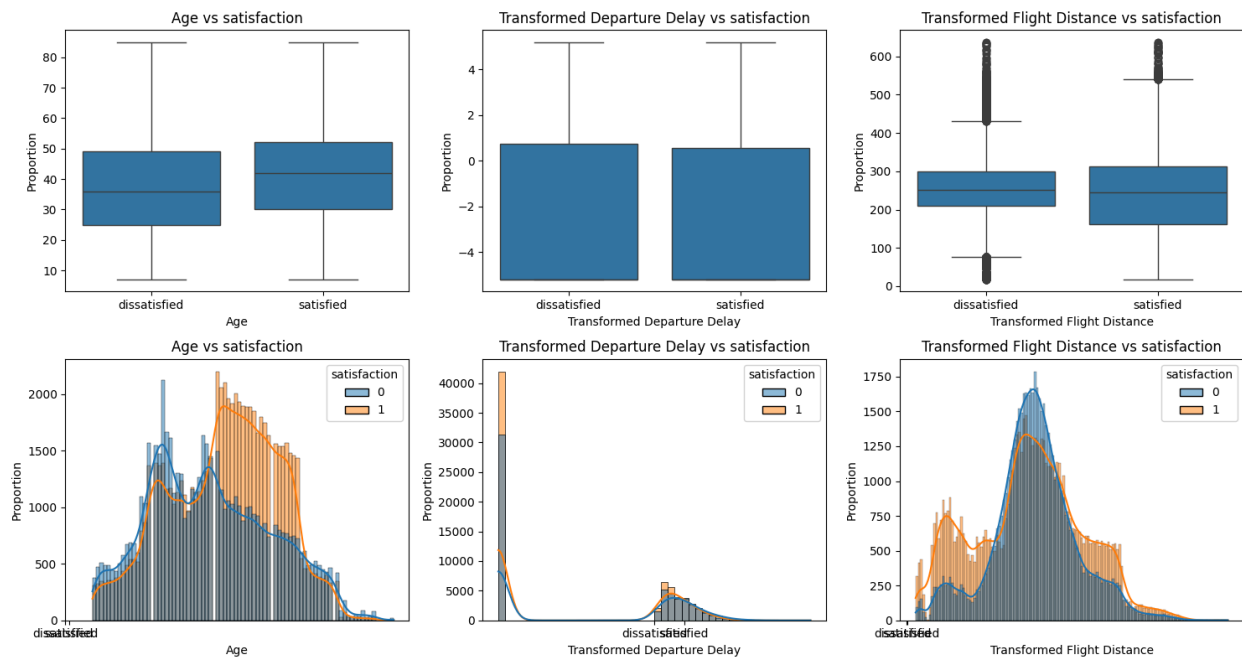


Figure 3. Boxplots and histograms (with KDE) of numerical variables grouped by target variable after transformation

Categorical and Ordinal

Categorical variables in the dataset include *Gender*, *Customer Type*, *Type of Travel*, and *Class*. The frequency of male (49.27%) and female (50.73%) customers are highly balanced. Closer examination shows that while the distribution of *satisfied* (44.01%) and *dissatisfied* (55.99%) male customers are balanced, there is an imbalance in the level of satisfaction among female customers (satisfied: 65.14%; dissatisfied: 34.86%).

Customer Type is divided into two categories where there is a high disparity between the number of loyal (81.70%) and disloyal (18.31%) customers. Notably, a larger proportion of loyal customers are satisfied (61.63%) compared to those who are dissatisfied (38.37%). Disloyal

customers exhibit the opposite trend, with satisfied individuals (23.97%) representing a smaller share than their dissatisfied counterparts (76.03%). This may suggest that the benefits of being a loyal customer has a positive effect on customer satisfaction and this categorical variable may have importance in predicting satisfaction.

Although a notable imbalance is present within the distribution of the *Type of Travel* feature, the level of customer satisfaction remains highly balanced across each travel category.

For the *Class* feature, *Eco* (58,309) and *Business* (62,160) have significantly higher numbers than customers in *Eco Plus* (9,411). However, both *Eco* and *Eco Plus* have similar satisfaction distributions which are slightly balanced, Business class customers have an overwhelming number of satisfied customers with 70.64%. We can infer from this information that seating in Business class may also have an influence in predicting customer satisfaction.

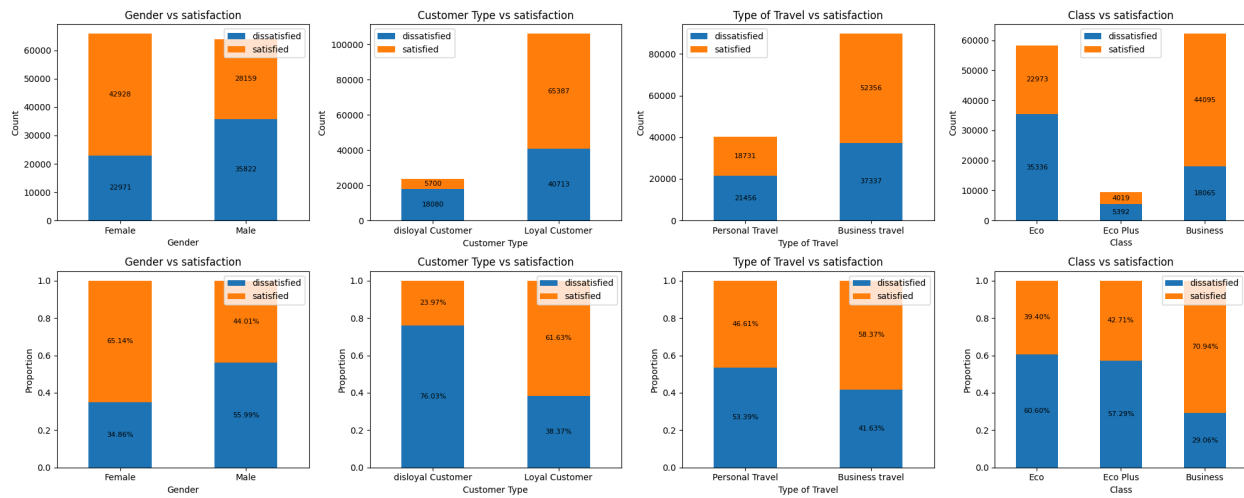


Figure 4. Stacked barplots of categorical variables grouped by target variable

Since none of the ordinal features have an absolute correlation coefficient larger than 0.8, we will keep all of them for now and will be investigated further (see Figure 5).

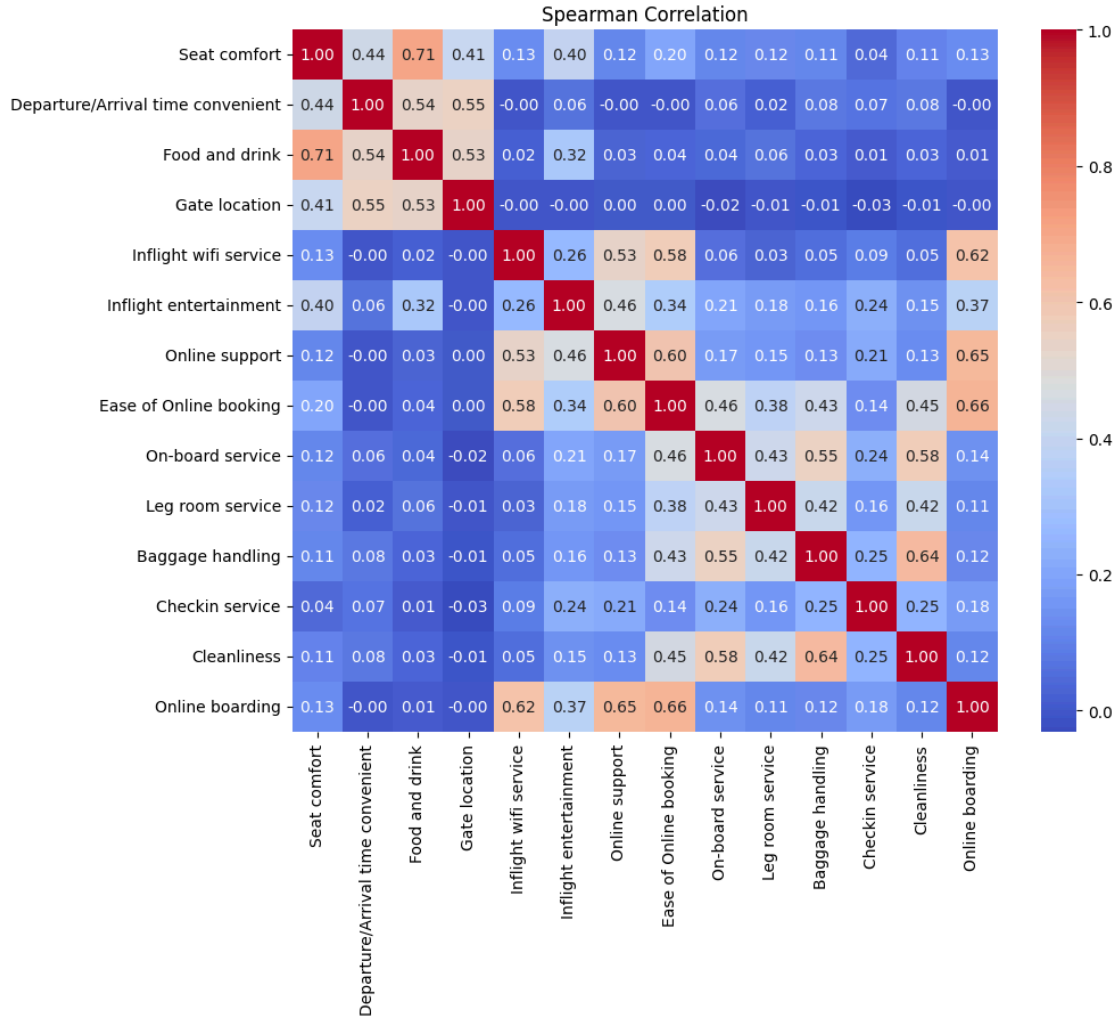


Figure 5. Spearman correlation heatmap of Likert scale variables

Feature Selection

In selecting features that we will model for predictive analysis, we first engineer features that we think are helpful in building a more robust model. Then, we remove features that provide minimal information about the target feature using Mutual Information. Next, we evaluate the predictor variables for high correlation and eliminate one from each pair to reduce redundancy. We will also explore tree-based feature importance to rank the variables based on their predictive power and ensure that the final model is both accurate and interpretable. It should be noted that since part of the study's objective is to identify attributes that highly influence customer satisfaction, predictors with Likert scale values are retained as much as possible unless it is deemed insignificant by the aforementioned metrics.

Engineered Features

From our Exploratory Data Analysis, we identified specific patterns suggesting that seating class is a significant driver of customer satisfaction. To better capture these relationships and potentially improve model performance, we engineered a new feature: *In Business Class*, a binary indicator derived from the *Class* variable. Instead of dividing the customers between Economy, Economy Plus, and Business class, this will be a variable indicating whether they are in Business class or not. As a consequence, the *Class* feature will be excluded henceforth.

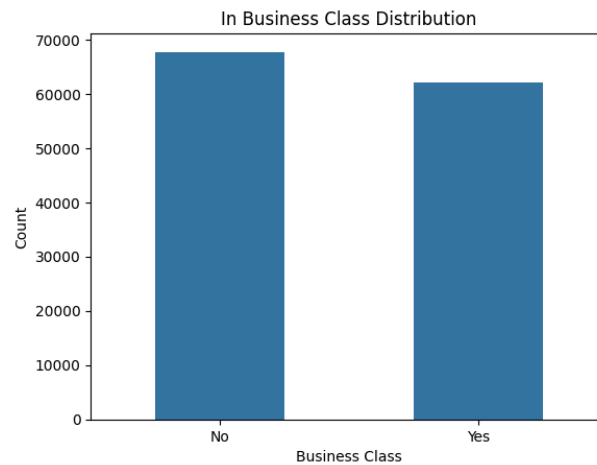


Figure 6. In Business class distribution

Mutual Information

Mutual information scores help us filter out noise in the dataset by calculating the statistical dependence between each predictor variable and the target variable, satisfaction. Based on the calculated scores presented in Table 3, we establish a threshold of 0.02 to identify and remove weak predictors. Consequently, features such as *Gate location* (0.011654), *Transformed Departure Delay* (0.007430), and *Departure/Arrival time convenient* (0.003929) are dropped from further analysis due to their negligible mutual information with the target. We retain the remaining variables, including high-scoring features like *Inflight entertainment* (0.237109), *Ease of Online booking* (0.113404), *Seat comfort* (0.143333), and *Type of Travel* (0.018140), for deeper investigation and subsequent model training.

Feature	Mutual information score	Feature	Mutual information score
Inflight	0.235100	Checkin service	0.046077

entertainment			
Seat comfort	0.142397	Food and drink	0.038378
Ease of Online booking	0.111713	Inflight wifi service	0.035550
Online support	0.100630	Gender	0.031445
On-board service	0.071129	Transformed Flight Distance	0.026129
Online boarding	0.066401	Age	0.024921
Business Class	0.061589	Type of Travel	0.018140
Leg room service	0.061265	Gate location	0.015080
Customer Type	0.055863	Transformed Departure Delay	0.006089
Baggage handling	0.054720	Departure/Arrival time convenient	0.002741
Cleanliness	0.053809		

Table 3. Mutual information scores of each variable

Correlation Analysis

For each type of feature, there are no highly correlated variables. Specifically, an evaluation of the ordinal, numerical, and categorical features indicates that none exhibit an absolute correlation coefficient larger than 0.8. Thus, all variables are retained for further investigation, and no features are dropped at this stage of the analysis.

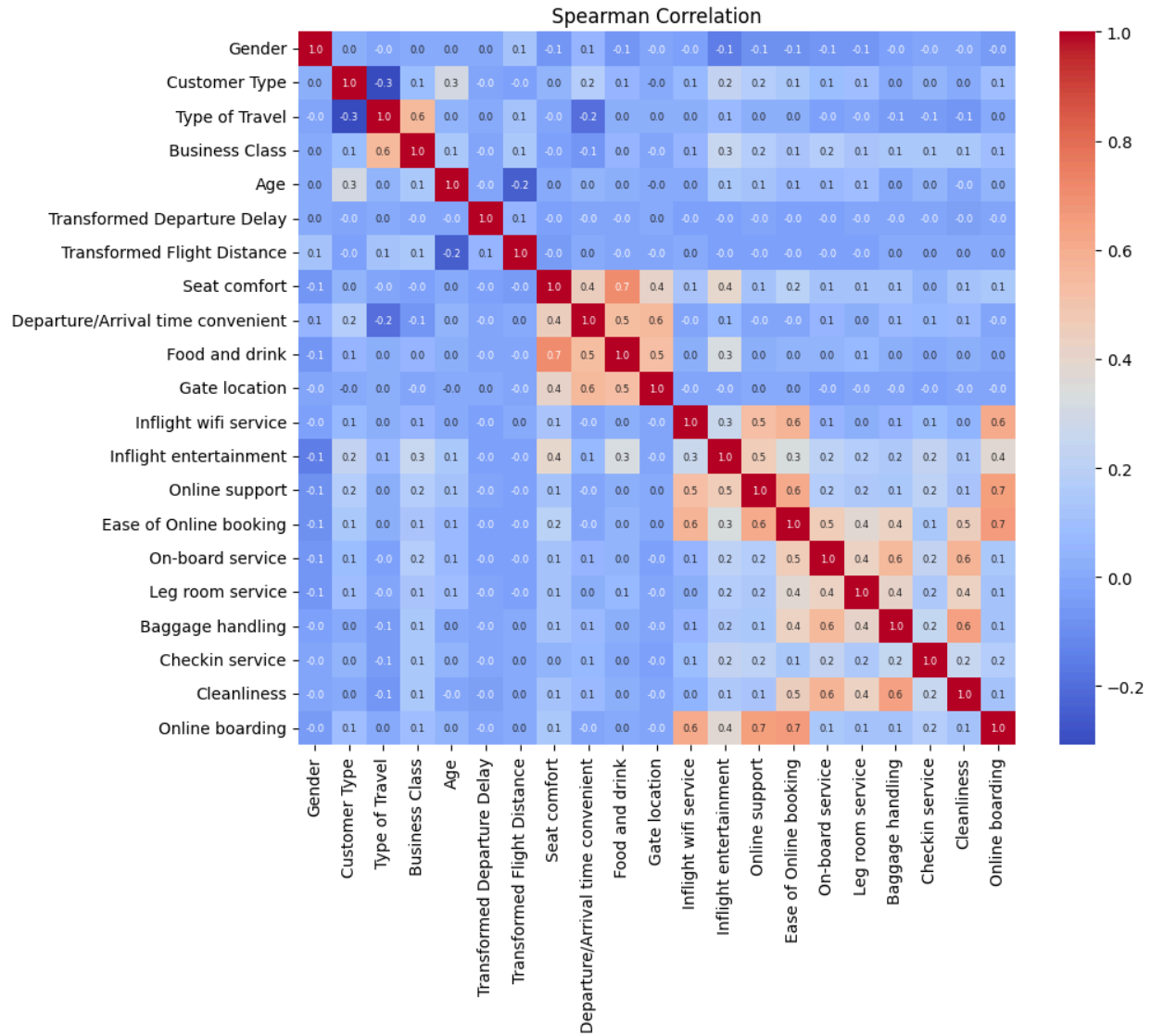


Figure 7. Spearman correlation heatmap of all possible predictors

Tree-Based Feature Importance

Tree-based algorithms, including Random Forest and LightGBM provide a robust mechanism for assessing feature importance by analyzing how much each predictor contributes to reducing variance or impurity across the construction of decision trees. During model training, these algorithms calculate the total decrease in node impurity attributed to each feature, providing a clear ranking of which variables most strongly influence the target outcome, *satisfaction*. By aggregating these importance scores, we can identify and select the most

impactful predictors, ensuring that our final model focuses on the variables that drive meaningful patterns in the dataset, while discarding those that contribute little to predictive power.

We used a Borda Count system in choosing our predictors for modeling. First, we rank our features based on their MI score, Random Forest importance, LightGBM importance, and XGBoost importance. Then, we sum the ranks for each feature (e.g., a feature ranked 2nd, 1st, and 4th gets a score of 7). Finally, select the features with the lowest total rank scores resulting in Table 4.

From our result, we will keep features in the top 12 of the Borda Count system which mainly includes Likert scale variables such as *Inflight entertainment*, *Seat comfort*, and *Ease of Online booking*. The categorical features *Customer Type* and *In Business Class* that we expected to have importance during the EDA are also in the top 12. *Transformed Flight Distance*, *Age*, *Gender*, and *Inflight wifi service* were excluded from the final feature set because they consistently exhibited relatively low importance across Mutual Information, Random Forest, and LightGBM, indicating limited contribution to predicting customer satisfaction compared with the other variables. Although *Cleanliness* also received a relatively low overall ranking, it was retained because further analysis demonstrated a clear association between *Cleanliness* ratings and customer satisfaction, and it achieved a moderate ranking in Mutual Information. Its lower importance in the tree-based models is likely attributable to shared predictive information with other service-related features rather than a lack of relevance. Therefore, *Cleanliness* was retained to preserve a meaningful service quality indicator in the final model.

Feature	Borda Count	Feature	Borda Count
Inflight entertainment	3	Online boarding	33
Seat comfort	6	Checkin service	34
Ease of Online booking	9	Baggage handling	35
Online support	16	Transformed Flight Distance	37
Leg room service	20	Age	39
Customer Type	23	Gender	40

In Business Class	23	Cleanliness	40
On-board service	24	Inflight wifi service	48
Food and drink	29		

Table 4. Borda count system results

Methodology

Train-test Split and Cross Validation

To train and evaluate the models, the dataset was partitioned into training and testing sets using a 4:1 split. By setting the stratification parameter to 'True', we guaranteed that the training and testing subsets precisely mirrored the class label distribution of the full dataset. Furthermore, a 5-Fold Stratified Cross Validation was carried out on the selected algorithms using their optimal configurations.

Models Evaluated

Logistic Regression

Logistic Regression is used to establish a simple and interpretable baseline classification model against which more complex machine learning algorithms can be compared. It is widely used in classification problems because it estimates the probability of class membership while remaining computationally efficient. Although it assumes a linear relationship between the predictors and the log-odds of the outcome, it provides an important benchmark for evaluating whether more sophisticated algorithms offer meaningful performance improvements.

Random Forest

Random Forest improves predictive performance by reducing overfitting commonly observed in single decision trees. This algorithm combines multiple decision trees through bootstrap aggregation, producing a more stable and accurate model (Brieman 2001). It also provides estimates of feature importance, which are valuable for identifying the factors that most strongly influence customer satisfaction.

LightGBM

LightGBM is used to assess a high-performance gradient boosting algorithm designed for large-scale datasets such as the one used in this study. It employs a histogram-based learning and leaf-wise tree growth, enabling faster training and lower memory consumption than many

traditional boosting methods (Ke et al., 2017). Its efficiency makes it perfect for our dataset while maintaining strong predictive performance.

Hyperparameter Tuning

To enhance model performance, hyperparameter optimization was conducted using RandomizedSearchCV. For the Random Forest Classifier, a 3-fold stratified cross-validation strategy was implemented, executing 30 iterations to explore the search space defined in Table 5. The tuning process was guided by the *'f1_weighted'* metric to ensure a balanced evaluation across classes.

Similarly, the LightGBM Classifier was optimized using RandomizedSearchCV with a 3-fold stratified cross-validation approach. This process also involved 30 iterations to explore the parameters outlined in Table 6, also utilizing the *'f1_weighted'* score as the primary evaluation metric to determine the optimal configuration.

Parameter	Values
n_estimators	randint(100, 500)
max_depth	[10, 20, 30, None]
min_samples_split	randint(2, 15)
min_samples_leaf	randint(1, 10)
max_features	['sqrt', 'log2', None]
bootstrap	[True, False]

Table 5. Random Forest Classifier search space for randomized search

Parameter	Values
n_estimators	randint(100, 600)
learning_rate	uniform(0.01, 0.19)
max_depth	randint(3, 15)
num_leaves	randint(20, 150)
min_child_samples	randint(10, 100)

ssubsample	uniform(0.6, 0.4)
colsample_bytree	uniform(0.6, 0.4)
reg_alpha	uniform(0, 5)
reg_lambda	uniform(0, 5)

Table 6. LightGBM Classifier search space for randomized search

Model Evaluation

Model Evaluation Metrics

- **Accuracy:** Represents the overall correctness of the model, calculated as the proportion of correctly predicted observations over the total number of observations.
- **Precision:** The proportion of predicted positive observations that are actually correct, determining how reliable the positive predictions are.
- **Recall:** The proportion of actual positive observations that are correctly identified by the model, measuring its ability to capture all positive instances (also referred to as the True Positive Rate).
- **F1-score:** The harmonic mean of precision and recall, providing a single balanced metric that is particularly useful for evaluating classification performance.
- **ROC-AUC:** Represents the Area Under the Receiver Operating Characteristic curve. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. TPR is the proportion of actual positives correctly identified, while FPR is the proportion of actual negatives incorrectly identified as positives. The AUC value measures the model's overall ability to distinguish between classes, where a score of 0.5 indicates random performance and 1.0 represents perfect classification.

Model Comparison

Based on the performance metrics presented in Table 7, the Random Forest model is the overall best performer. While both ensemble methods—Random Forest and LightGBM—significantly outperformed the Logistic Regression baseline, Random Forest achieved higher scores across almost all key metrics:

- **Accuracy:** Random Forest (0.941022) is slightly higher than LightGBM (0.935363).

- **Precision and F1-score:** Random Forest shows higher reliability in positive predictions (0.948709 precision) and a stronger harmonic balance between precision and recall (0.945965 F1-score) compared to LightGBM.
- **ROC-AUC:** While LightGBM holds a very slight edge in its ability to distinguish between classes (0.986333 vs. 0.984956), Random Forest's consistent lead in accuracy, precision, and F1-score makes it the most robust model for this specific classification task.

Logistic Regression remains the least effective model across all evaluated metrics, confirming that more sophisticated tree-based algorithms were necessary to capture the complexities of the dataset.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.820219	0.831516	0.842161	0.836805	0.895982
Random Forest	0.941022	0.948709	0.943237	0.945965	0.984956
LightGBM	0.935363	0.943478	0.938102	0.940782	0.986333

Table 7. Performance metrics for each model

Confusion Matrix and ROC Curve

A confusion matrix is a table that serves as a fundamental tool for evaluating the accuracy of a classification model by comparing predicted class labels against the actual ground truth. It breaks down the model's predictions into four categories: True Positives (correctly identified positive cases), True Negatives (correctly identified negative cases), False Positives (negative cases incorrectly predicted as positive), and False Negatives (positive cases incorrectly predicted as negative). This matrix provides deep insight into specific types of errors the model makes, allowing for a more nuanced understanding of precision, recall, and overall classification performance beyond a simple accuracy score.

The Receiver Operating Characteristic (ROC) curve is a graphical representation used to evaluate the performance of a binary classification model at various threshold settings. It plots the True Positive Rate (TPR), which represents the model's ability to correctly identify positive instances, against the False Positive Rate (FPR), which indicates the proportion of negative

instances incorrectly classified as positive. The Area Under the Curve (AUC) provides a single metric to summarize this performance, where a score closer to 1.0 indicates a superior ability to distinguish between classes, while 0.5 represents performance no better than random guessing.

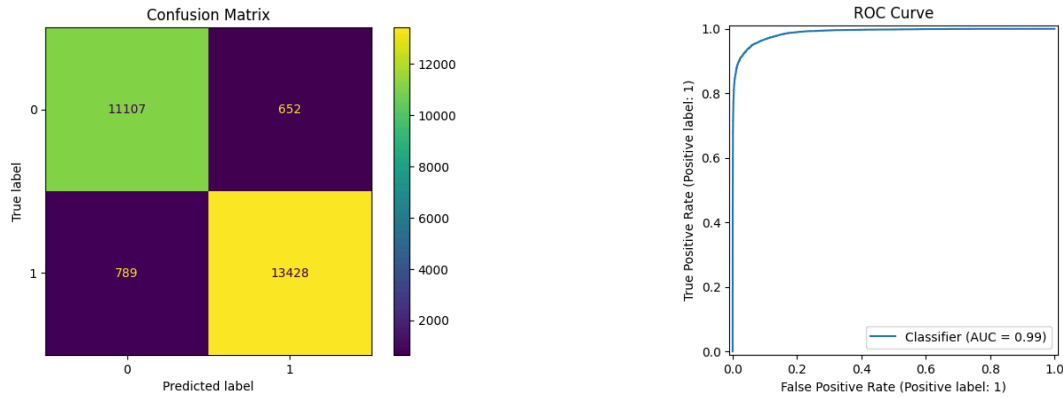


Figure 8. (a) Confusion matrix and (b) ROC Curve of the best Random Forest model

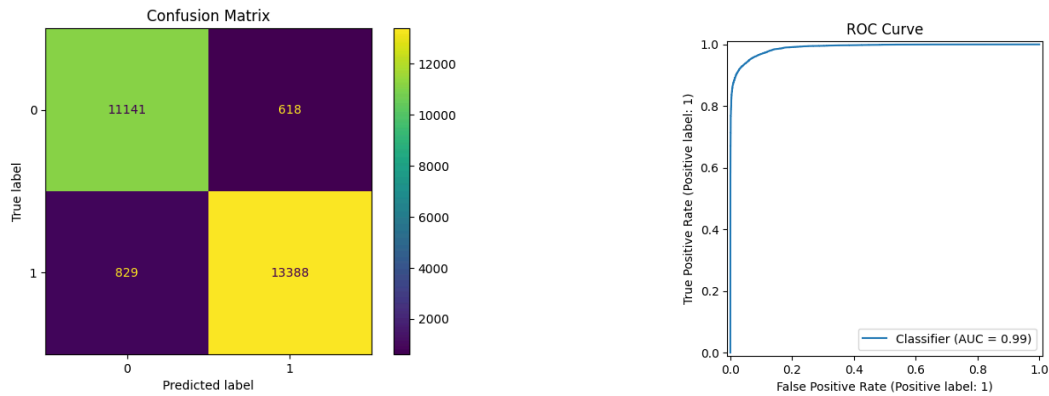


Figure 9. (a) Confusion matrix and (b) ROC Curve of the best LightGBM model

Figures 8 and 9 visualize the classification efficacy of the Random Forest and LightGBM models through their respective confusion matrices and ROC curves. Both models exhibit robust performance, with the confusion matrices revealing a high concentration of correct predictions along the diagonal, confirming the strong accuracy metrics previously noted. The ROC curves for both algorithms further illustrate this effectiveness, as they are prominently positioned toward the top-left corner, signifying high sensitivity and a strong ability to distinguish between satisfied and dissatisfied passengers. While the models demonstrate slightly different metrics, these visual representations collectively validate that both tree-based ensemble methods achieve superior discriminative power compared to the baseline model.

Despite the performance metrics achieved by the Random Forest model, the LightGBM classifier was selected as the final model for this study. This decision was driven by its superior performance during hyperparameter tuning, where LightGBM achieved a higher cross-validation score. This result demonstrates that LightGBM provides better generalization and robustness, making it the more reliable choice for predicting airline customer satisfaction in a real-world environment.

Furthermore, while the Random Forest model achieves the highest overall performance metrics, relying solely on these aggregate figures can overlook the specific business impact of different error types. In the context of airline customer satisfaction, minimizing false positives—instances where the model incorrectly predicts a passenger as "satisfied" when they are actually "dissatisfied"—is strategically critical. A higher false positive rate creates a dangerous false sense of security, potentially leading the airline to overlook passengers who require service interventions. By prioritizing a model with a lower false positive rate, the airline ensures that it more accurately captures truly dissatisfied customers, allowing for targeted service recovery and effectively mitigating the risk of customer churn that might otherwise go unnoticed.

Model Interpretation

Feature Importance

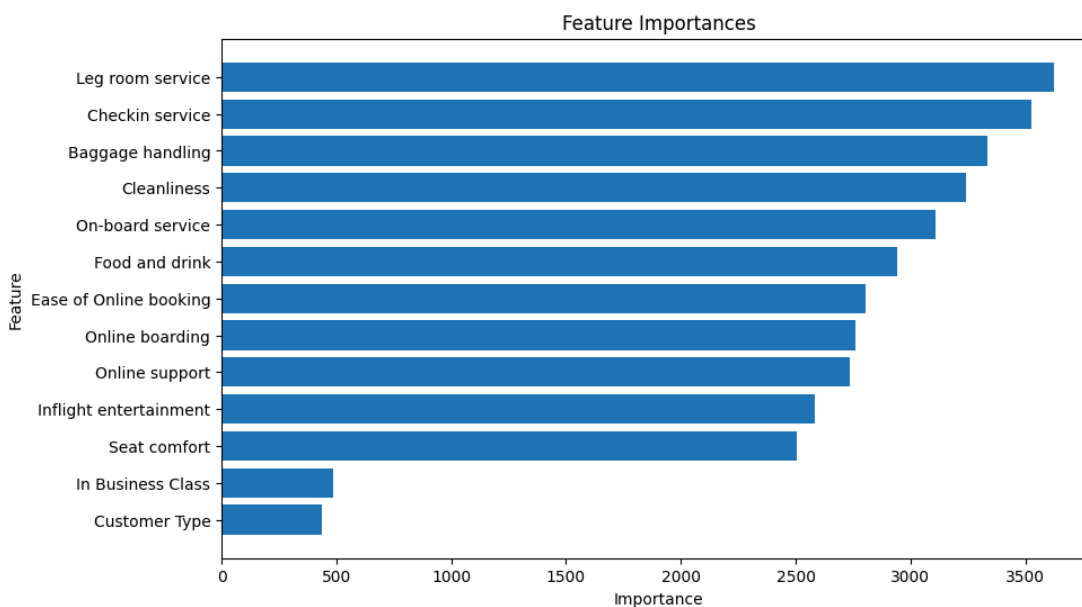


Figure 10. Feature importance in the best LightGBM model

Feature importance analysis provides critical insights into the primary determinants of airline customer satisfaction. As visualized in Figure 10, the model indicates that service attributes related to inflight experience and pre-flight digital touchpoints exert the strongest influence on predictive outcomes. Specifically, features such as seat comfort, inflight entertainment, and ease of online booking emerge as robust drivers of customer satisfaction. While tree-based importance highlights the significant contribution of operational metrics like leg room and check-in services, the SHAP interpretation confirms that positive ratings across core passenger experiences—particularly in comfort and entertainment—are the most reliable indicators of a satisfied passenger, whereas inconsistencies or low scores in these areas consistently push the model toward predicting dissatisfaction.

Interestingly, the features *In Business Class* and *Customer Type* exhibit relatively low importance scores in the final model, despite initial trends observed during Exploratory Data Analysis. This discrepancy suggests that while these high-level categorical factors are associated with satisfaction, specific operational service attributes—such as seating comfort and digital touchpoint quality—are much more powerful and direct drivers of passenger feedback. Ultimately, the model indicates that the quality of the actual service delivery carries significantly more weight in determining customer satisfaction than broad passenger-level demographic or booking class information.

SHAP Analysis

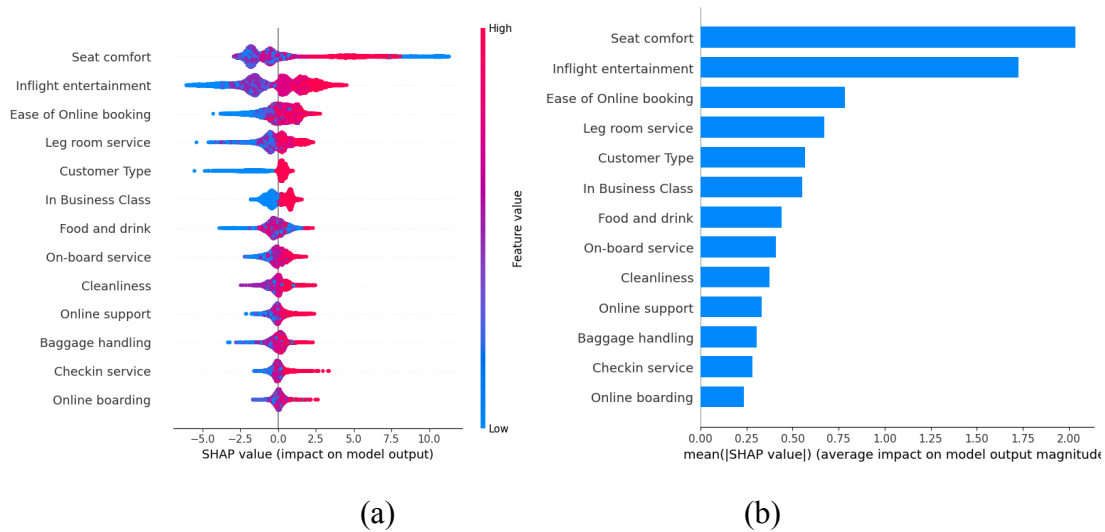


Figure 11. (a) Dot plot or violin-style scatter plot and (b) Bar plot of the summary of SHAP values

Global Interpretation

Global SHAP analysis, as supported by the Feature Importances in Figure 10, reveals that overall customer satisfaction is primarily driven by inflight service features and digital booking convenience. These inferences are derived by analyzing SHAP value magnitudes and directions, which quantify the contribution of each feature to the model's final prediction for the dataset as a whole. Features such as *Inflight entertainment*, *Ease of Online booking*, and *Seat comfort* exhibit the strongest positive impact on model predictions, consistently increasing the likelihood of a passenger being classified as satisfied when rated favorably. Conversely, low scores in core passenger touchpoints—particularly online services and seating quality—substantially push predictions toward dissatisfaction, demonstrating that customer satisfaction is highly sensitive to the quality of these core operational attributes.

Local Interpretation

Local SHAP explanations provide itemized attribution for individual passenger predictions, showing how specific feature combinations drive a particular outcome. These inferences are derived by analyzing SHAP value magnitudes and directions, which quantify the contribution of each feature to the model's final prediction for individual passengers. For example, for a satisfied passenger in Business Class, high ratings in *Inflight entertainment* and *Ease of Online booking* act as the primary positive contributors that push the log-odds prediction above the decision threshold. In contrast, for misclassified or dissatisfied instances, local force plots highlight specific pain points—such as low *Seat comfort* or poor digital support—that heavily penalize the prediction score despite positive ratings in secondary attributes.

Error Analysis

	Correct Mean	Misclassified Mean
Inflight entertainment	3.359248	3.682101
Seat comfort	2.788944	3.636489
Ease of Online booking	3.484162	3.301313
Online support	3.534306	3.334485
Leg room service	3.496840	3.342778
In Business Class	0.488361	0.281963

Customer Type	0.819561	0.765031
On-board service	3.474092	3.323428
Food and drink	2.807330	3.545266
Online boarding	3.365078	3.254319
Checkin service	3.334706	3.276434
Baggage handling	3.705981	3.598480
Cleanliness	3.712544	3.619212
Actual	0.545803	0.572909
Predicted	0.545803	0.427091

Table 8. Comparison of feature means between correctly and misclassified observations

An in-depth analysis of misclassified observations provides key insights into the operational boundaries and decision mechanics of the predictive model. By evaluating instances where predicted customer satisfaction deviates from actual feedback, specific passenger profiles and systematic error patterns can be identified.

1. Which customers are difficult to classify?

- **Passengers with Mixed or Conflicting Service Ratings:** Customers who report highly disparate ratings across touchpoints—such as rating inflight service and seat comfort exceptionally high while giving low scores to digital booking or online support—present substantial classification difficulty. The tree-based models struggle to establish a definitive decision boundary when core attributes yield conflicting signals.
- **Disloyal Passengers in Economy Class:** Non-loyal customers (Customer Type mean of 0.765 in misclassified cases vs. 0.820 in correctly classified cases) traveling in Economy class display significantly higher variability in overall satisfaction. Their overall rating is more sensitive to minor service discrepancies, making their feedback harder to predict reliably.
- **Borderline Rating Profiles:** Passengers whose ratings cluster near the mid-point (ratings of 3 or 4 across multiple Likert-scale attributes) generate lower model confidence, as slight shifts in individual ratings can tip the predicted class between satisfied and dissatisfied.

2. Are there systematic errors?

- False-Positive Bias Driven by Premium Class Status:** A systematic error pattern emerges among passengers who travel in Business Class or receive high ratings on primary inflight attributes (e.g., higher misclassified means for Seat Comfort at 3.636 vs 2.789, and Food and Drink at 3.545 vs 2.807). The model tends to over-index on these positive premium attributes, incorrectly predicting satisfaction even when severe dissatisfaction is recorded in pre-flight digital services.
- Underestimation of Digital Touchpoint Friction:** The feature mean comparison shows that misclassified observations exhibit lower scores in Ease of Online Booking (3.301 vs 3.484) and Online Support (3.334 vs 3.534). The model systematically underestimates the degree to which poor pre-flight digital experiences drag down overall passenger satisfaction despite otherwise satisfactory flight conditions.

Discussion and Conclusion

An examination of the misclassified instances reveals specific patterns regarding which passenger profiles present classification challenges. Passengers with conflicting service ratings—such as evaluating *Seat comfort* or *Inflight entertainment* exceptionally high while scoring digital touchpoints like *Ease of Online booking* low—account for a significant portion of prediction errors. Additionally, non-loyal passengers traveling in economy class exhibit higher variance in reported satisfaction, making their overall experience harder for tree-based models to distinctly boundary.

Systematic errors are predominantly concentrated in false-positive predictions among passengers with borderline ratings across core service attributes. In these cases, the presence of premium attributes (such as being *In Business Class*) occasionally overrides negative feedback in digital support attributes, leading the model to incorrectly classify a dissatisfied passenger as satisfied.

The empirical findings underscore the critical role that both inflight service quality and pre-flight digital touchpoints play in driving customer satisfaction. Across all models evaluated, attributes associated with inflight entertainment, seat comfort, and digital booking interfaces consistently emerged as the primary determinants of positive passenger feedback. These results align with established aviation and service quality literature (Chang & Yeh, 2002; Park et al.,

2006; Giao, 2017), which emphasizes that customer perceptions are formed across a continuum of touchpoints—from initial digital reservation to on-board comfort.

In summary, while the Random Forest classifier demonstrated slightly higher raw performance metrics during initial testing (Accuracy: 0.941, F1-score: 0.946), the LightGBM classifier was selected as the final operational model due to its superior cross-validation performance during hyperparameter tuning and lower false-positive rate, ensuring robust real-world generalization. The key drivers of passenger satisfaction identified include *Inflight entertainment*, *Seat comfort*, *Ease of Online booking*, *Online support*, and seating class position.

Recommendation

Business Recommendations

- **Enhance Digital Touchpoints:** Invest in streamlining online booking and support platforms to remove friction during the pre-flight phase, as ease of booking is a heavy contributor to overall satisfaction.
- **Prioritize Inflight Entertainment and Comfort:** Upgrade cabin seating and expand onboard media offerings, which represent the top global drivers of passenger satisfaction across both economy and premium classes.
- **Targeted Loyalty Interventions:** Establish rapid service recovery protocols specifically targeting disloyal or economy passengers who report low digital support scores, mitigating churn before dissatisfaction fully sets in.

Technical Recommendations

- **Incorporate Behavioral and Temporal Data:** Future modeling efforts should integrate additional feature types, such as historical booking frequency, interaction timestamps, and customer service response times, to better resolve borderline classifications.
- **Explore Advanced Ensemble or Neural Architectures:** Experiment with deep neural network architectures or multi-model stacking to improve predictive capability on complex, mixed-rating customer profiles.
- **Establish Automated Retraining Pipelines:** Implement continuous monitoring and periodic model retraining on new customer feedback batches to adapt to shifting passenger expectations and seasonal travel variations.

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